Multiple Disease Predictions using Machine Learning and Deep Learning Algorithms

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***Abstract*— Machine studying and synthetic brain play an necessary position today. They are everywhere from self-driving automobiles to the clinical field. The healthcare enterprise generates giant quantities of affected person information that can be processed in a range of ways. Therefore, we use computing device gaining knowledge of to improve predictive structures that can pick out multiple illnesses simultaneously. Many current structures can solely predict one ailment at a time, with low accuracy. Poor accuracy can damage patient's health. Four of his illnesses are presently being investigated: a coronary heart tumor, a talent tumor, a lung tumor and diabetes, with the opportunity of including extra illnesses in the future. The person have to enter a sequence of sickness symptoms and the gadget determines whether or not the individual has the ailment or not. This effort may want to gain many humans as it would permit them to display their fitness and take steps to enhance their survival.**

***Keywords*— Heart disease, Brain Tumor, Diabetes, SVM, CNN, Bi-LSTM**

I INTRODUCTION

Data is a resource in modern Healthcare datasets helps to get main resources of patient details. In this paper, a generic architecture for illness prediction in the healthcare business is suggested [1]. Several current models concentrate, and also for COVID.

There is no one system that identifies several diseases at a time [2]. As a result , we are focusing on providing patients Hence, using Flask, we propose a solution for predicting numerous illnesses. We will evaluate Cardiovascular disease , Brain tumor classification ,and Diabetes illness analysis in this system. Several more illnesses can be added later. We will utilize machine learning methods , deep learning methods and Flask API (Application Programming Interface) to construct several illness prediction systems. The model can be safely used as a Pickle format [3].

The significance of this model discussion is that in diagnosing illnesses, characteristics that produce the condition are included, allowing for more efficient and accurate disease detection. The behavior of the completed model will be stored as a python pickle file. There will be also upcoming discussions about this study : Part II shows the survey of previous papers which was done. Part III explains the theme of project and it's explanation . Part IV briefs the outcome of the project with related pictorial results. Part V gives the inferences from the project.

II LITERATURE SURVEY

From the article [5], diabetes is the most hazardous disease in the world , causing a wide range of ailments such as blindness . In this article , scientists employed machine learning approaches to detect diabetic disease

since it is simple and adaptable to predict whether a patient is unwell. The goal of the investigation was to create a useful model for patient diagnose diabetic illness with reliable findings. They examined the accuracy of four key algorithms, from that Decision Tree which has the highest accuracy of 85%.

The major goal of the study is, because the heart is crucial in living creatures. As a result, the prediction and prognosis of disorders should be exact and correct since it is highly important and might cause death instances connected to heart [6].

From above reviewed articles, many machine learning models for healthcare analysis are designed to focus on one specific disease for each analysis. Unfortunately, there is no universal system that allows for multi-disorder diagnosis. Furthermore, several models exist with low accuracy, which can have a negative impact on patient's health outcomes. Organizations that wish to analyze their patient's health reports are forced to produce multiple models and leads to increased costs and time expenditures. In addition, some of these existing systems only consider a limited number of parameters, which can produce false or inaccurate results.

We regularly see sufferers die due to the fact they did no longer obtain the remedy as planned. Health structures do now not have time to figure out which sufferers to deal with first. On the one hand, the clinical enterprise produces a giant quantity of affected person fitness data. Detailed facts can be extracted from this data. So, I determined to begin a assignment to predict a range of illnesses the use of this records and superior computer studying techniques.

Our project, which combines three computing device getting to know models , objectives to become aware of sufferers with coronary heart disease , and diabetes early in order to deal with these at excessive threat of positive ailments first. . For our project, we first apprehend the trouble and determine what form of records the mission requires. We accrued three distinct datasets for three Kaggle computing device studying models. After accumulating the data , we carried out splendid information evaluation and visualization for higher understanding.

III PROPOSED SYSTEM

With this concept of prediction, it is now able to diagnose many diseases simultaneously, thereby eliminating the need for users to visit different sites for each prediction. For example, we have developed a system that can predict four diseases at once, namely Heart, Brain Tumor, and Diabetes disease. These three diseases are known to be correlated to each other, making multi-disease prediction especially relevant for healthcare analysis.

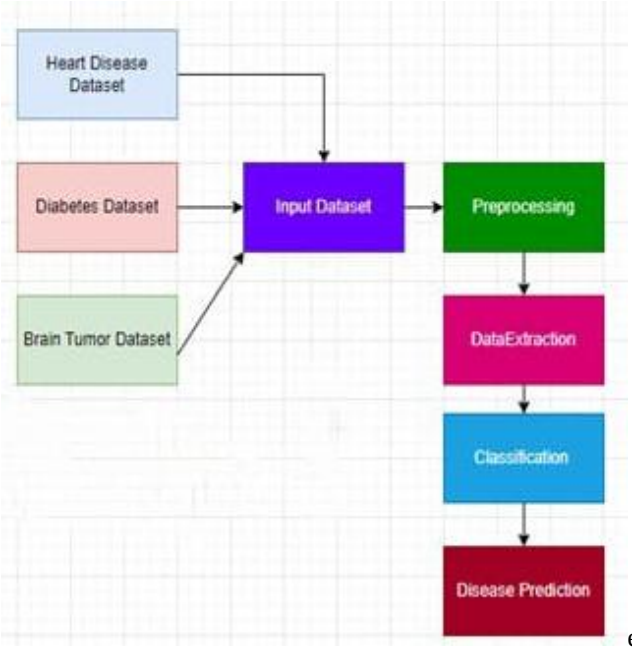

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Figure 1. System Architecture

We will look each stage in detail from upcoming sections.

A. Data Collection

The dataset of cardiac disease is utilized for modelling training and evaluation. The Cleveland dataset from the UCI repository was accessible through the open-source site "Kaggle". 13 functions, 13 records, and one column of targets make up this dataset. The target column has two classes: binary values with heart disease and without heart disease [8]. For Brain tumors there are nearly 3200 and 4600 images respectively. Diabetes dataset is collected from "Kaggle" repository.

B. Data Preprocessing

The quality and quantity of data play a crucial role in the accuracy and reliability of the predictions made by the system. Pre-processing the data ensures that the data used to train the machine learning algorithms is of high quality and completeness, which helps to increase the accuracy of the disease prediction system.

Moreover, data pre-processing also involves data normalization, feature selection, and feature extraction, which helps to improve the quality of the data [9]. By normalizing the data, the disease prediction system can handle variables with different scales and units, allowing for better analysis and prediction. Feature selection and feature extraction reduce the number of

variables used in the analysis, which can improve the efficiency and accuracy of the machine learning algorithms.

C. Feature Selection

In machine learning models, the performance of the prediction model can be improved by selecting the features creates most impact on the target feature. This reduces the dimensionality of the input data, which can enhance the model's performance, reduce overfitting, and reduce the computational complexity [10]. Correlation method is one of the commonly used feature selection techniques in disease prediction systems. In feature selection, correlation is used to identify the bonding between the independent features (predictor feature) and the dependent feature (target feature) [11].

The correlation heatmap of Brain tumor model in Figure 2 below:

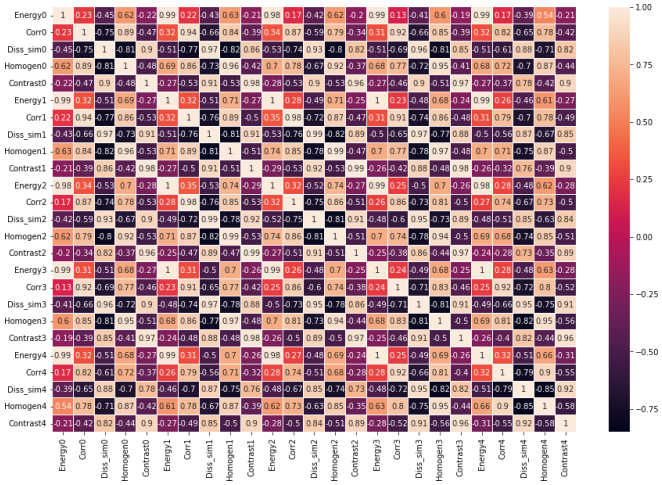

Figure 2. correlation heatmap of Brain Tumor Model

The above heatmap gives us 5 important features to extract image data which was used in model training model.

D. Classification

Once the correlation has done, the features get extracted from the dataset for heart and diabetes and from images for brain tumor. We have classified the data into training set as 75% and testing set as 25%. Training set undergoes algorithm implementation with SVM, Logistic and Linear Regression, CNN, Bi-LSTM, and testing set used to examine the performance of pre-trained model [12].

IV ALGORITHM AND IMPLEMENTATION

A vastly used and most was also used to assess the highly complexed data and to manage it.

A. SUPPORT VECTOR MACHINES(SVM)

This is used to categorize the datasets in different dimensions by identifying hyperplanes that can separate them. There are different hyperplanes that can be used to split the data into two sets. The SVM algorithm aims to find the hyperplane with the largest margin between the closest points of different classes. By maximizing the margin, the algorithm increases its ability to accurately classify new data points [14].

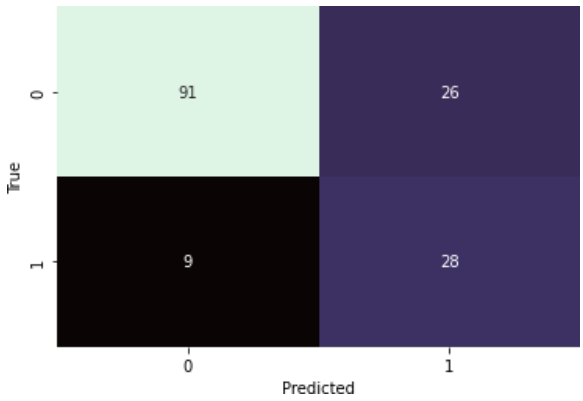

Figure 3. confusion matrix of SVM (Diabetes)

B. LOGISTIC REGRESSION

This is a analytical method investigates The technique uses a logistic function, which is the collective distribution function of the logistic distribution, to calculate the possibility of the dependent variable [15].

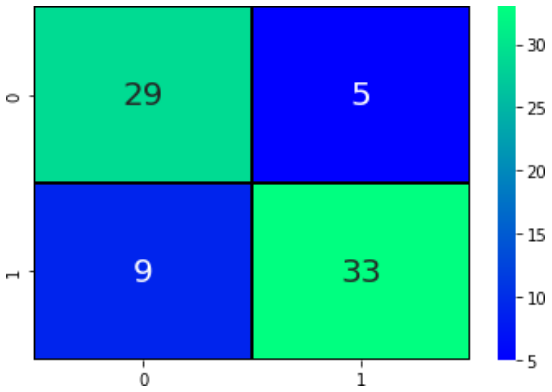

Figure 4. confusion matrix of logistic regression (Heart)

C. CONVOLUTIONAL NEURAL NETWORK(CNN)

In this method, the input image is first passed through a series of convolutional layers, features are derived from input image using filters. Each filter scans the image and generates a feature map that highlights different patterns or edges. The resulting feature maps are then processed through pooling layers minimizes the dimensionality of

the feature maps while preserving the most important information. The resulting features are then given into fully connected layers, which classify the input image into different categories. During training, the network adjusts the weights of the filters to minimize the difference between the predicted output and the ground truth labels [16].

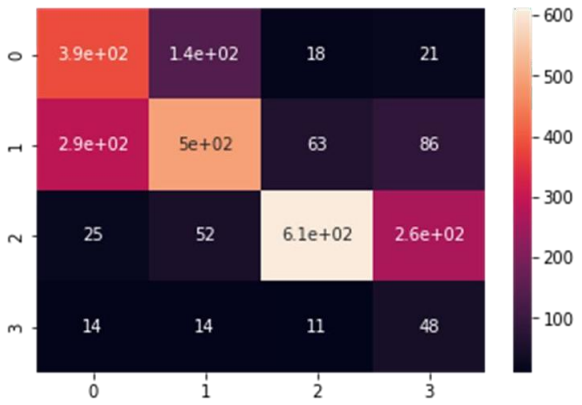

Figure 5. confusion matrix of CNN (Brain Tumor)

It summarizes predictions made by model on testing dataset, and compares them with the true labels of the data. The matrix contains four values: and true negatives (TN) [18]. These values can be used to calculate various metrics that The importance of confusion matrix in result analysis is that it provides detailed viewpoint of the performance of the model beyond a single metric such as accuracy. It shows how the model works in both right and wrong situations, and can help identify specific areas of improvement for the model on and which classes it is not, and adjust the model accordingly.

Here are the formulas for some of the commonly used metrics that can be derived from a confusion matrix [19]:

1. Accuracy: It is a value of right outcomes out of overall predictions made by the model and provides complete measure of model's performance.

   Accuracy = (TP + TN) / (TP + TN + FP + FN)

2. Sensitivity (also called Recall or True Positive Rate): It is a value of right examples that are correctly identified by the system. It measures the model's ability to identify positive examples.

   Sensitivity = TP / (TP + FN)

3. Specificity: It is a value of negative examples that are correctly identified by model. It calculates the system ability to identify negative examples.

   Specificity = TN / (TN + FP)

4. Precision: It is the value of correctly predicted values that are right. It measures the model's ability to make accurate positive predictions.

   Precision = TP / (TP + FP)

Each of these metrics provides a different perspective on performance of system and can be helpful for analyzing its strengths and weaknesses [20]. By using these metrics in conjunction with the confusion matrix,.

V RESULTS

Performance metrics for four models were calculated by applying various methodologies as shown below:

A. ACCURACY FOR HEART DISEASE:

When we use the C – hyperparameter in the SVM model, the model performs more precisely with the 56-regularization value than other values of C [21]. From the functionality of above tuning, the performance metrics resulted as shown below:

Algorithm	Accuracy	Sensitivity	Specificity	Precision
Logistic Regression	90%	95%	87%	80%
Decision Tree	84%	83%	84%	77%
Random Forest	84%	83%	84%	77%
Support Vector Machine	94%	96%	90%	85%

Table-1: heart disease

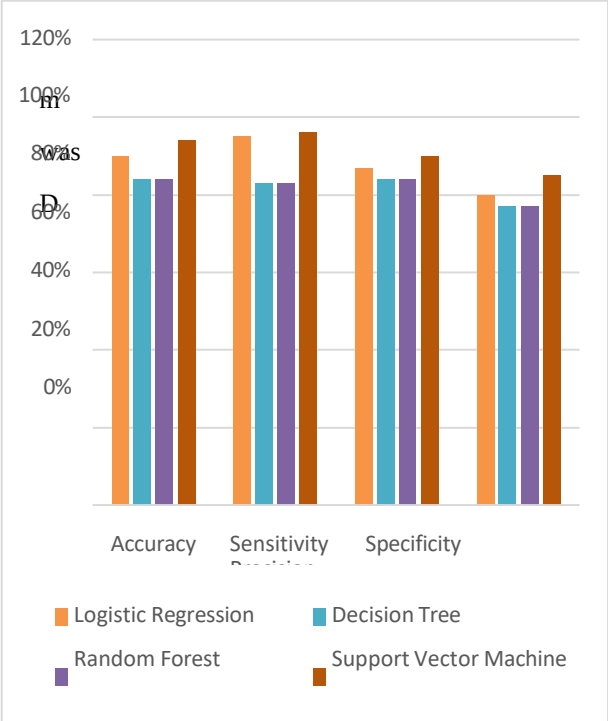

Fig 6: Heart disease Accuracy

B. ACCURACY FOR BRAIN TUMOR CLASSIFICATION:

ALGORITHM	CNN-BiLSTM	CNN	LSTM
ACCURACY	96	93.78	89.67
SENSITIVITY	91	87.11	78.34
SPECIFICITY	95	95.68	93.15
POSITIVE PREDICTIVE VALUE	91	88.39	79.52
NEGATIVE PREDICTIVE VALUE	95	95.77	92.91

Table-2: brain tumor

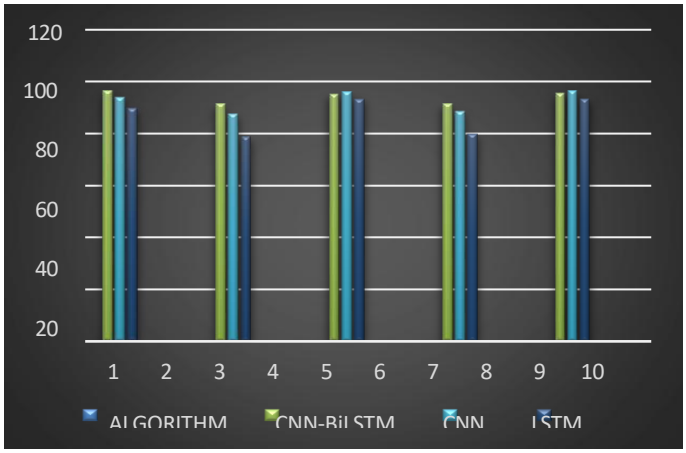

Fig 7:Brain Tumor Accuracy

From the above-mentioned values of performance metrics, a final graphical representation

developed to understand about accuracy of each model.

C.ACCURACY FOR DIABETES:

ALGORITHM	ACCURACY	SENSTIVITY	SPECIFICITY	PRECISION
LOGISTIC REGRESSION	94%	93%	87%	94%
RANDOM FOREST	98%	95%	90%	95%

Table-3: diabetes disease

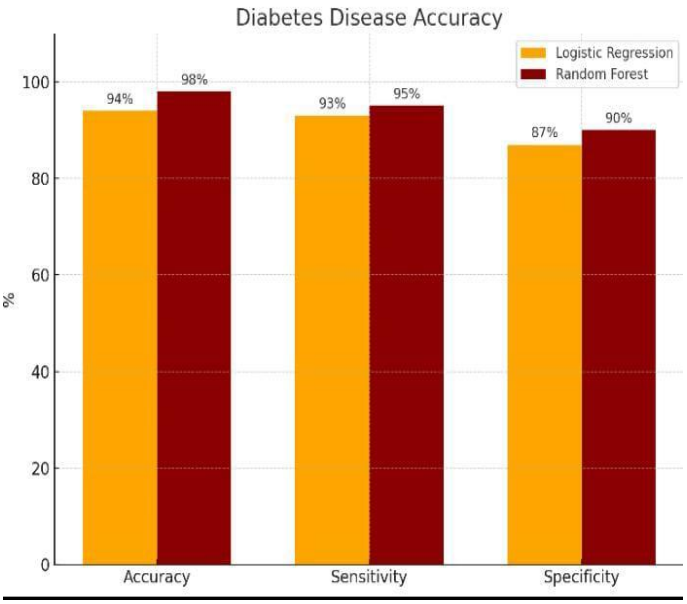

Fig 8: Diabetes disease Accuracy

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V CONCLUSION

The primary intention of the lookup is to increase a device that can precisely predict a large vary of diseases. Thanks to this project, customers no longer want to navigate between their more than one websites, which additionally saves time.

Early analysis of the sickness can extend lifestyles and keep away from monetary difficulties . We use a aggregate of computing device getting to know and deep gaining knowledge of algorithms such as SVM, logistic regression, selection trees, random forest, CNN, to obtain most accuracy.

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